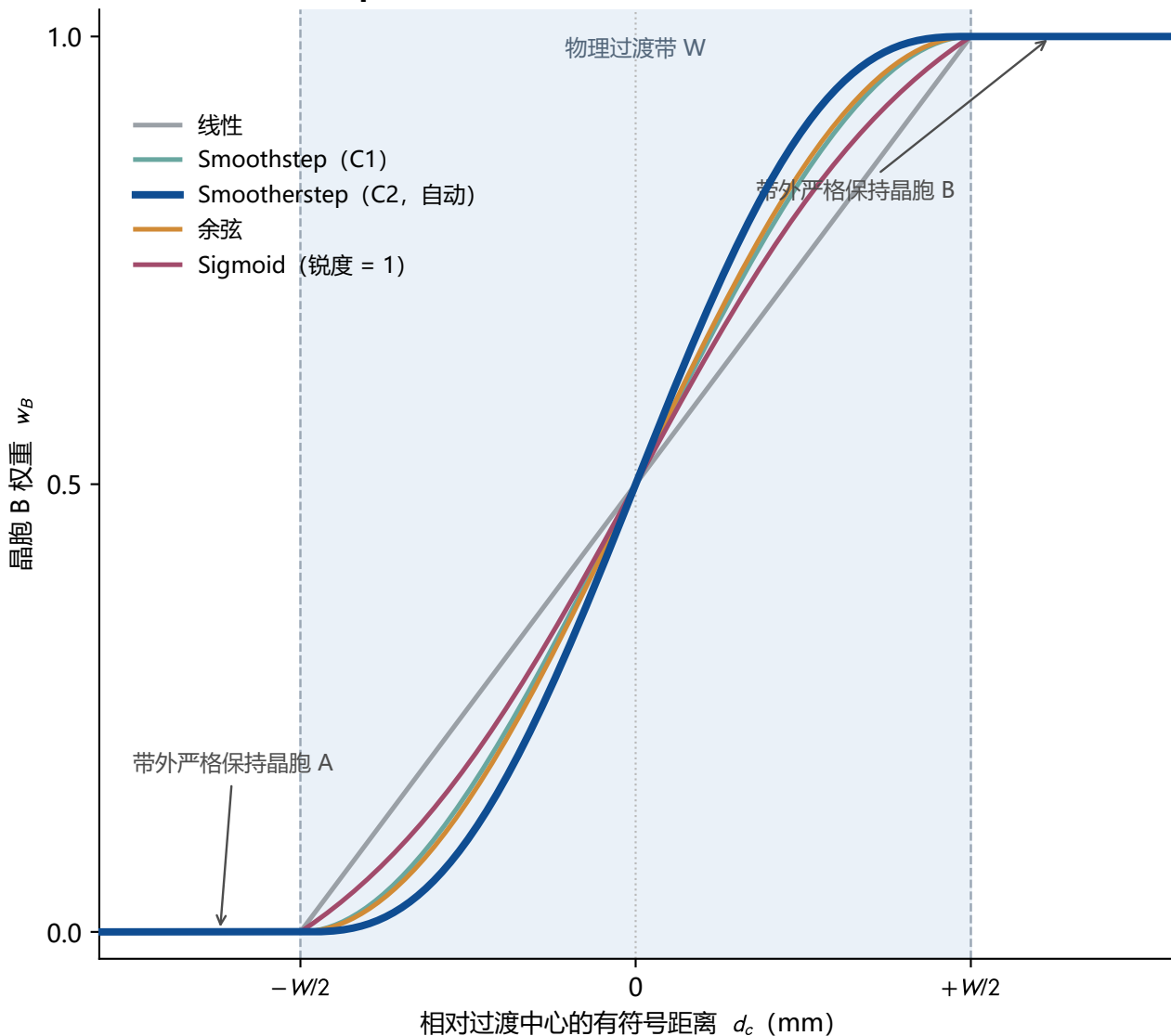


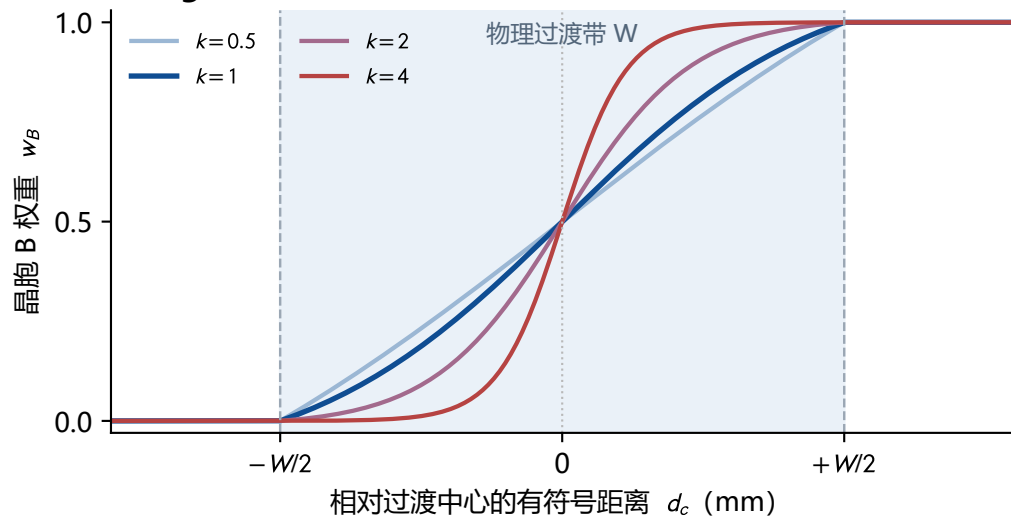
晶格过渡的空间 Ramp 权重与隐式场融合

过渡宽度限定作用范围；权重曲线控制两类晶胞在带内的连续交接方式

a 项目支持的 Ramp 权重函数



b Sigmoid 锐度 (不改变 W)



c 项目权重与基础场融合公式

$$d_c = (\mathbf{x} - \mathbf{p}) \cdot \hat{\mathbf{n}} - \delta, \quad t = \text{clip}\left(\frac{d_c + W/2}{W}, 0, 1\right)$$

$$w_B = R(t), \quad w_A = 1 - w_B, \quad R_{\text{auto}}(t) = 6t^5 - 15t^4 + 10t^3$$

$$R_{\text{sig}}(t) = \frac{\sigma[\kappa(2t - 1)] - \sigma(-\kappa)}{\sigma(\kappa) - \sigma(-\kappa)}, \quad \kappa = \ln(9)k$$

$$F_{\text{base}} = (1 - w_B)F_A + w_B F_B, \quad F_{\text{final}} = \max(F_{\text{domain}}, F_{\text{tr}})$$

F_{tr} 为基础融合场经可选局部配准与拓扑约束修正后的过渡场